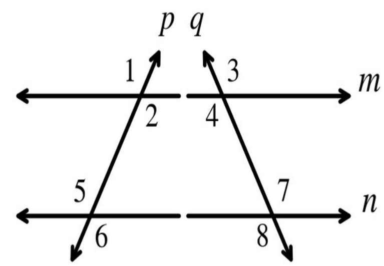
SC Honors Geometry - Nappier
3.3 – Proving Lines Parallel • Homework

Name: ____________________________________

Use the first diagram. Which two lines (if any) must be parallel? Name the theorem.



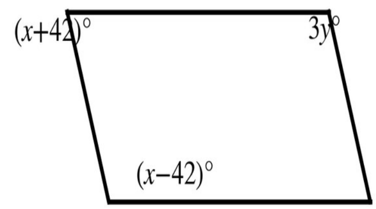
1. $\angle 3 \cong \angle 7$

2. $\angle 4 \cong \angle 7$

3. $\angle 2 \cong \angle 5$

4. $\angle 6$ and $\angle 8$ are supplementary

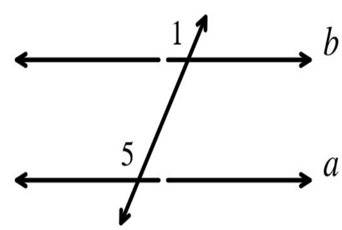
5. $\angle 2 \cong \angle 7$



Complete each statement and solve.

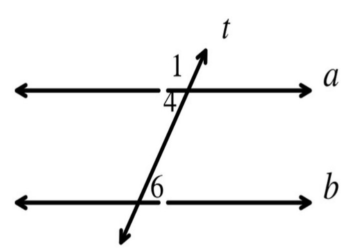
6. If two lines are cut by a transversal so that a pair of consecutive interior angles are ______________, then the lines are parallel.

7. Find the values of x and y that make the opposite sides of the second figure parallel.



8. In the third diagram, $m\angle 1 = (4x + 7)°$ and $m\angle 5 = (6x - 19)°$. Find the value of x that makes $a \parallel b$.

9. Is it possible for two lines cut by a transversal to have congruent corresponding angles and NOT be parallel? Explain.



Complete the two-column proof. (Use the last diagram.)

10. Given: $\angle 1$ and $\angle 6$ are supplementary. **Prove:** $a \parallel b$

Statements	Reasons
1. $\angle 1$ and $\angle 6$ are supplementary	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**